Course Syllabus: Software Engineering 1st Semester, A.Y. 2026–2027

Course Information

Instructor: [SE Instructor Name] | Units: 3 | Schedule: TTh 10:30 AM–12:00 NN | Room: IT Lab 1

1. Course Description

Software Engineering introduces students to the systematic development of high-quality software. The course covers software process models, requirements engineering, software design principles, design patterns, software testing strategies, project management, software quality assurance, and DevOps practices. Students will apply these principles in a group software development project that spans the entire semester, culminating in a working software prototype and a formal project documentation package.

2. Learning Outcomes

LO 1: Apply software process models (Agile, Scrum, Waterfall) to plan and manage a software development project.

LO 2: Write clear and testable software requirements using use cases and user stories.

LO 3: Apply object-oriented design principles (SOLID) and common design patterns to software architecture.

LO 4: Design and execute a test plan covering unit, integration, and system testing.

LO 5: Use version control (Git/GitHub) and CI/CD tools in a team software project.

3. Weekly Schedule

Week	Topic	Assessment
1	Introduction to Software Engineering; SDLC Models	Quiz 1
2	Agile Methodology; Scrum Framework; Sprint Planning	Lab 1: Scrum Board Setup
3	Requirements Engineering; User Stories; Acceptance Criteria	Lab 2: Requirements Doc
4	Software Architecture; Layered, MVC, Microservices	PT 1: Architecture Diagram
5	Object-Oriented Design; SOLID Principles	Quiz 2
6	Design Patterns: Singleton, Factory, Observer, Strategy	Lab 3: Design Pattern Implementation
7	Version Control with Git; Branching Strategies; Pull Requests	Lab 4: GitHub Workflow
8	Midterm Examination	Midterm Exam
9	Software Testing; Unit Testing with pytest or JUnit	PT 2: Test Suite

10	Integration Testing; System Testing; UAT	Lab 5
11	Software Quality Assurance; Code Reviews; Static Analysis	Quiz 3
12	DevOps Basics; CI/CD Pipelines with GitHub Actions	PT 3: CI/CD Pipeline
13	Software Project Management; Estimation; Risk Management	Lecture
14	Documentation; API Documentation; README Best Practices	Lab 6
15	Group Project Presentations: Working Software Demo	Group Demo
16	Final Examination	Final Exam

4. Grading System

Component	Weight
Quizzes / Participation	10%
Laboratory Exercises	20%
Performance Tasks	20%
Group Software Project	20%
Midterm Exam	15%
Final Exam	15%

5. Group Project Requirements

Each group (3–4 members) must develop a working software application addressing a real-world problem. The project must use Git for version control with a shared repository, maintain a product backlog, complete at least 3 sprint cycles, and submit a final project documentation package including SRS, design documents, test plan, and user manual.

Technology Stack: Students may use any programming language and framework approved by the instructor. Web-based applications (React, Django, FastAPI) and mobile applications (Flutter, React Native) are both acceptable.

6. References

1. Sommerville, I. (2016). Software Engineering (10th ed.). Pearson.

2. Martin, R.C. (2017). Clean Architecture. Prentice Hall.

3. Gamma, E. et al. (1994). Design Patterns: Elements of Reusable Object-Oriented Software. Addison-Wesley.

4. GitHub Documentation. https://docs.github.com

5. Atlassian Agile Coach. https://www.atlassian.com/agile